8 Averages of x^t

8.1 Evaluation of integrals

Any integral of the form

$$\int (A + Bt)^{\alpha} \cdots (M + Nt)^{\mu}\, dt, \tag{1}$$

where A, B, α, $\ldots$, M, N, μ are constants, can be evaluated in terms of R functions. The integral in general will not be well defined if one of the linear functions in parentheses, say $A + Bt$, vanishes at an interior point of the path of integration, but it may vanish at an endpoint if $\operatorname{Re} \alpha > -1$. A factor which vanishes at an endpoint will be treated separately from the rest. If 2α, $\ldots$, 2μ are integers, the integral is elementary, elliptic, or hyperelliptic, and the explicit formulas given below will be useful for elliptic integrals in Chapter 9. Many integrals with trigonometric or hyperbolic integrands are connected with (1) by substitutions such as $t = \sin\theta$ or $t = \cosh^2\theta$ [see Nellis (1966, Table I)]. We shall denote a line segment by $[x, y]$ as in (7.1-10).

FORMULA 8.1-1 Let $x, y \in \mathbb{C}$ and $z, w, b \in \mathbb{C}^k$. Let $a, a' \in \mathbb{C}_>$ and assume $a + a' = \sum_{i=1}^{k} b_i$. Assume further that $z_i + w_i t \neq 0$ for every $t \in [x, y]$ and $i = 1, \ldots, k$, and let $(z + wy)/(z + wx)$ denote the k-tuple with ith component $(z_i + w_i y)/(z_i + w_i x)$. If the path of integration is straight, then

$$\int_x^y (t-x)^{a-1}(y-t)^{a'-1}\prod_{i=1}^k (z_i+w_i t)^{-b_i}\,dt$$
$$= \mathrm{B}(a,a')(y-x)^{a+a'-1}\prod_{i=1}^k (z_i+w_i x)^{-b_i}R_{-a}\left(b,\frac{z+wy}{z+wx}\right)$$
$$= \mathrm{B}(a,a')(y-x)^{a+a'-1}\prod_{i=1}^k (z_i+w_i y)^{-b_i}R_{-a'}\left(b,\frac{z+wx}{z+wy}\right), \quad (2)$$

where $\mathrm{ph}(t-x)=\mathrm{ph}(y-x)=\mathrm{ph}(y-t)$ and $\mathrm{ph}(z_i+w_i t)$ is continuous in t on $[x,y]$ for $i=1,\ldots,k$.

Remark An integral of this form which does not satisfy the condition $a+a'=\sum_{i=1}^k b_i$ can always be made to do so by inserting a suitable power of $1+0t=1$. For example the integrand $(t-x)(z_1+w_1 t)^{-1/2}$ is to be regarded as $(t-x)(y-t)^0(z_1+w_1t)^{-1/2}(1+0t)^{-5/2}$.

Proof Set $t=(1-u)x+uy$, whence

$$z_i+w_i t=(z_i+w_i x)[1-u+u(z_i+w_i y)/(z_i+w_i x)].$$

The middle member of (2) follows from (6.8-2) and the last member from (6.8-15). □

Similar formulas follow for the cases in which one limit of integration is infinite. The path of integration is again straight but is now assumed for convenience to be parallel to the real axis if not part of it. This can always be arranged if necessary by rotating the plane.

FORMULA 8.1-2 Let $x\in\mathbb{C}$ and $z,w,b\in\mathbb{C}^k$. Let $a,a'\in\mathbb{C}_>$ and assume $a+a'=\sum_{i=1}^k b_i$. For $i=1,\ldots,k$ assume further that $w_i\neq 0$ and that $z_i+w_i t\neq 0$ for every t such that $t-x\in\mathbb{R}_+$. Let $x+zw^{-1}$ denote the k-tuple with ith component $x+z_iw_i^{-1}$. Then

$$\int_x^\infty (t-x)^{a'-1}\prod_{i=1}^k (z_i+w_i t)^{-b_i}\,dt = \mathrm{B}(a,a')\prod_{i=1}^k w_i^{-b_i}R_{-a}(b,x+zw^{-1}), \quad (3)$$

where $\mathrm{ph}(t-x)=0$ and $\mathrm{ph}(z_i+w_i t)$ is continuous in t and tends to $\mathrm{ph}\,w_i$ as $t\to\infty$ for $i=1,\ldots,k$.

Proof As t describes the path of integration, $t+z_iw_i^{-1}$ describes a ray with endpoint $x+z_iw_i^{-1}$ parallel to the positive real axis. Since $t+z_iw_i^{-1}=(z_i+w_it)w_i^{-1}\neq 0$, the ray does not contain the origin and $\mathrm{ph}(t+z_iw_i^{-1})=\mathrm{ph}(z_i+w_it)-\mathrm{ph}\,w_i$ tends continuously to 0 as $t\to\infty$. Hence $|\mathrm{ph}(t+z_iw_i^{-1})|<\pi$. Set $t=\tau+x$ and use (6.8-6). □

FORMULA 8.1-3 Let $x\in\mathbb{C}$ and $z,w,b\in\mathbb{C}^k$. Let $a,a'\in\mathbb{C}_>$ and assume $a+a'=\sum_{i=1}^k b_i$. For $i=1,\ldots,k$ assume further that $w_i\neq 0$ and

$z_i - w_i t \neq 0$ for every t such that $x - t \in \mathbb{R}_+$. Let $zw^{-1} - x$ denote the k-tuple with ith component $z_i w_i^{-1} - x$. Then

$$\int_{-\infty}^{x} (x-t)^{a'-1} \prod_{i=1}^{k} (z_i - w_i t)^{-b_i}\, dt = \mathrm{B}(a, a') \prod_{i=1}^{k} w_i^{-b_i} R_{-a}(b, zw^{-1} - x), \tag{4}$$

where $\mathrm{ph}(x - t) = 0$ and $\mathrm{ph}(z_i - w_i t)$ is continuous in t and tends to ph w_i as $t \to -\infty$ for $i = 1, \ldots, k$.

Proof Set $t = -t'$ and $x = -x'$, and use (3). ▯

An integral over the whole real line can be broken into two parts and evaluated by (3) and (4), but sometimes other methods are effective (as in Exercise 8.1-4).

8.2 The Schwarz–Christoffel mapping and elliptic functions

A function w which is holomorphic on an open set Ω in the z plane maps Ω onto an open set in the w plane. If $w' \neq 0$ on Ω, the mapping is called conformal because it preserves the size and orientation of angles: at any fixed point $z_0 \in \Omega$, the equation $dw = w'(z_0)\, dz$ implies that $\mathrm{ph}(dw) - \mathrm{ph}(dz)$ is independent of dz. The Schwarz–Christoffel mapping is a conformal mapping of the open upper half z plane onto the inner region of a polygon in the w plane. As z describes the upper edge of a cut along the real axis in the direction of increasing real values, w describes the closed polygon in the positive direction. The vertices of the polygon are images of chosen points on the extended real line, and the angles at these vertices are parameters of the mapping.

Let $x \in \mathbb{R}^k$, $k \geq 2$, and assume that the components of x are distinct. Let $b \in \mathbb{R}^k$, define a by $a + 1 = \sum_{i=1}^{k} b_i$, and assume $0 < a \leq 1$ and $0 < b_i < 1$ for $i = 1, \ldots, k$. Consider the differential equation

$$dw = -a \prod_{i=1}^{k} (z - x_i)^{-b_i}\, dz, \tag{1}$$

where $0 < \mathrm{ph}(z - x_i) < \pi$ if $\mathrm{Im}\, z > 0$. Because dw/dz is holomorphic on the upper half-plane, so is the particular solution

$$w(z) = a \int_{z}^{\infty} \prod_{i=1}^{k} (t - x_i)^{-b_i}\, dt, \tag{2}$$

where the path of integration lies in the half-plane. We shall write $w(\infty) = 0$ to signify that $w(z) \to 0$ as $z \to \infty$ in any manner in the upper half-plane.

If the plane is cut along the real axis and z moves along the upper edge of the cut in the direction of increasing real values, then $\mathrm{ph}(dz) = 0$ while

$\mathrm{ph}(z - x_i)$ is π if $z < x_i$ and 0 if $z > x_i$. It follows that $\mathrm{ph}(dw)$ is constant except for a jump of $b_i \pi$ when z passes x_i for each $i = 1, \ldots, k$. Hence w describes a polygonal line which changes in direction by angles equal to the jumps in $\mathrm{ph}(dw)$. The line starts at the point $w(\infty) = 0$ when $z = -\infty$, its direction turns counterclockwise through a total angle of $\sum_{i=1}^k b_i \pi = (a+1)\pi$ as z traverses the real axis, and as $z \to +\infty$, it returns to $w(\infty) = 0$ to close the polygon. At this point the line turns through an angle $(1 - a)\pi$ to make a total change of 2π in $\mathrm{ph}(dw)$. Since $(1 - a)\pi$ is between 0 and π like the other angles, the polygon is convex and cannot intersect itself. Because $dw/dz \neq 0$ on the upper half-plane, the function w maps this half-plane (which lies to the left as z traverses the real axis) conformally onto the inner region of the polygon (which lies to the left as w traverses the polygon). [For a proof that the mapping is one-to-one and onto, see Markushevich (1965, Volume 2, p. 130).]

Evaluating (2) by (8.1-3),

$$w(z) = R_{-a}(b_1, \ldots, b_k; z - x_1, \ldots, z - x_k). \tag{3}$$

Because of the condition $a + 1 = \sum_{i=1}^k b_i$, or briefly $a' = 1$, the derivative of this R function is elementary by Exercise 5.9-18 and (6.6-5). We assumed the real numbers $x_1, \ldots, x_k$ to be distinct, but in view of Theorem 5.2-4 we may let $x_i = x_j$ provided $b_i + b_j < 1$. This corresponds to letting two vertices of the polygon coalesce provided the sum of their angles of turn is less than π. The angle of turn is sometimes called the outer angle; it is the difference between π and the inner angle at the same vertex of the polygon. The vertices of the polygon corresponding to (3) are $w(x_1), \ldots, w(x_k)$, and $w(\infty) = 0$, and their respective outer angles are $\beta_1 \pi, \ldots, \beta_k \pi$, and $(1 - a)\pi$. If $a = 1$, the polygon passes through 0 but does not have a vertex there. In order to locate the vertices we shall study in the next section the behavior of the R function as one variable tends to 0. It is clear from (2) that each $w(x_i)$ is finite because $b_i < 1$.

The following theorem is a summary for convenient reference.

THEOREM 8.2-1 Let $z \in \mathbb{C}$, $a \in \mathbb{R}$, and $b, x \in \mathbb{R}^k$, $k \geq 2$. Assume that the components of x are distinct and $\operatorname{Im} z > 0$. Let $z - x$ be the k-tuple with ith component $z - x_i$. Assume further that $0 < a \leq 1$, $a + 1 = \sum_{i=1}^k b_i$, and $0 < b_i < 1$ for $i = 1, \ldots, k$. Then the function

$$w(z) = R_{-a}(b, z - x) \tag{4}$$

maps the open upper half-plane conformally and one-to-one onto the inner region of a closed convex polygon. The inner angles of the polygon are $(1 - b_1)\pi, \ldots, (1 - b_k)\pi$, and $a\pi$. The corresponding vertices are $w(x_1), \ldots, w(x_k)$, and $w(\infty) = 0$, where $w(x_i)$ denotes the limit of $w(z)$ as $z \to x_i$ from the upper half-plane. As z describes the real axis in the direction of increasing real values, w describes the polygon in the positive direction, the last side of the polygon being an interval of the positive real axis ending at $w(\infty) = 0$.

Remark The polygon can be translated, rotated, and magnified by a further mapping $f(w) = Cw + D$, where C and D are complex constants.

EXAMPLE 8.2-2 (*Weierstrass' elliptic function*) A particular case of the Schwarz–Christoffel mapping has an inverse mapping which can be continued analytically to make a doubly periodic function holomorphic on $\mathbb{C}$ except for poles. Let x_1, x_2, x_3 be real and distinct, Im $z > 0$, and

$$\begin{aligned} u(z) &= R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; z - x_1, z - x_2, z - x_3) \\ &= \frac{1}{2}\int_z^\infty [(t - x_1)(t - x_2)(t - x_3)]^{-1/2}\, dt. \end{aligned} \tag{5}$$

The theory of elliptic functions developed from Abel's idea of regarding the limit of integration as a function of the integral (z as a function of u in the case at hand). Even if the x_i are complex, $z(u)$ can be proved with some difficulty to be a doubly periodic meromorphic function, but we shall deal only with the much simpler case of real x_i. Because (5) has fundamental importance in the subject of elliptic integrals, we introduce the abbreviation

$$R_F(x, y, z) = R_{-1/2}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; x, y, z), \tag{6}$$

where the subscript F comes from Legendre's notation for the elliptic integral of the first kind [see (9.2-11)]. Thus

$$u(z) = R_F(z - x_1, z - x_2, z - x_3). \tag{7}$$

By symmetry we may suppose $x_1 > x_2 > x_3$.

By Theorem 8.2-1 the function $u(z)$ maps the open upper half z plane conformally and one-to-one onto the interior of a rectangle in the u plane with vertices $u(x_1)$, $u(x_2)$, $u(x_3)$, and $u(\infty) = 0$. The inverse mapping $z(u)$ takes the interior of the rectangle in the u plane conformally and one-to-one onto the open upper half z plane. As u describes the rectangle in the positive direction, starting from $0 = u(\infty)$ and proceeding to $u(x_3)$, $u(x_2)$, $u(x_1)$, and back to 0 (see Fig. 8.2-1), z describes the real axis in the direction of increasing real values. Since it is real on the rectangle, $z(u)$ can be analytically continued outside the rectangle by Schwarz's reflection principle. This principle asserts that an analytic function with real boundary values on a line segment can be analytically continued across the segment by assigning conjugate complex values to the function at points related by reflection in the segment [see Titchmarsh (1939, Section 4.51)]. In the present

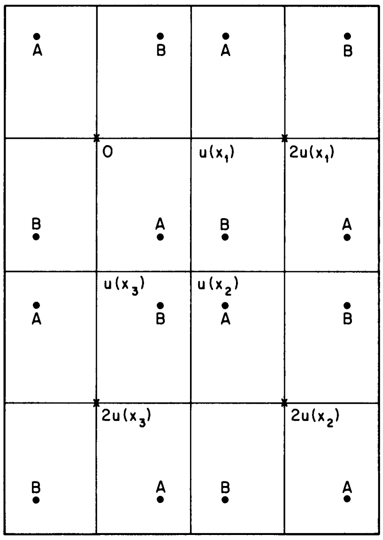

Figure 8.2-1 The u plane with $u(z) = R_F(z - x_1, z - x_2, z - x_3)$ and $x_1 > x_2 > x_3$. The lines are the image of the real axis of the z plane. The doubly periodic inverse function $z(u)$ takes the same value with positive imaginary part at all points labeled A and the complex conjugate value at all points labeled B, which are related to the points A by reflection in the lines. It takes real values on the lines and has a double pole at each point marked with a cross. The numbers $u(x_1)$, $u(x_3)$, and $u(x_2) = u(x_1) + u(x_3)$ are half-periods, $u(x_1)$ being real and positive and $u(x_3)$ purely imaginary.

case reflection in each side of the rectangle continues $z(u)$ analytically to the inner region of an adjacent rectangle. Since the continuation has real boundary values on each adjacent rectangle, the process can be repeated any number of times. Inside successive rectangles $z(u)$ takes alternately the values (with positive imaginary parts) inside the original rectangle and their complex conjugates (with negative imaginary parts). Thus $z(u)$ becomes a doubly periodic function, and the lengths of the sides of each rectangle are its half-periods. It is holomorphic on the u plane except for a double pole at $u = 0$ and at every point displaced from 0 by whole numbers of periods. The poles are double because (5) implies $u(z) \sim z^{-1/2}$ as $z \to \infty$ and $z(u) \sim u^{-2}$ as $u \to 0$. A doubly periodic meromorphic function is called an elliptic function. The single-valued elliptic function $z(u)$ is the function inverse to the many-valued elliptic integral $u(z)$.

By (5), (8.1-3), and (6.10-4), the real half-period is

$$u(x_1) = \frac{1}{2}\int_{x_1}^{\infty} [(t - x_1)(t - x_2)(t - x_3)]^{-1/2}\, dt = \frac{\pi}{2} R_K(x_1 - x_2, x_1 - x_3). \tag{8}$$

Since $x_1 > x_2 > x_3$ the path of integration for $u(x_3)$ lies in the upper half-plane [see (2)] and can be deformed by Cauchy's integral theorem into a real path from x_3 to $-\infty$. Since $u(x_3)$ lies on the negative imaginary axis (see Fig. 8.2-1),

$$\begin{aligned}u(x_3) &= \frac{1}{2}\int_{x_3}^{-\infty} [(t - x_1)(t - x_2)(t - x_3)]^{-1/2}\, dt \\ &= \frac{-i}{2}\int_{-\infty}^{x_3} [(x_1 - t)(x_2 - t)(x_3 - t)]^{-1/2}\, dt \\ &= -i\frac{\pi}{2} R_K(x_1 - x_3, x_2 - x_3). \qquad (9)\end{aligned}$$

In the last step we have used (8.1-4). Both half-periods are complete elliptic integrals of the first kind and can be computed numerically by the algorithm of the arithmetic–geometric mean (Example 6.10-2). The remaining vertex of the rectangle (see Fig. 8.2-1) is $u(x_2) = u(x_1) + u(x_3)$, also a half-period.

If $x_2 = 0$ and $x_3 = -x_1$, it follows from (8) and (9) that the rectangle is a square:

$$u(x_1) = \frac{\pi}{2} x_1^{-1/2} R_K(1, 2), \qquad u(x_3) = -iu(x_1). \qquad (10)$$

This is called the lemniscatic case because of its relation to Bernoulli's lemniscate (Exercise 8.3-7). Gauss discovered in 1799 that the perimeter of the lemniscate is an arithmetic–geometric mean and subsequently that any such mean is a period of a doubly periodic function.

By (1) the elliptic function $z(u)$ satisfies the nonlinear differential equation

$$dz/du = -2(z - x_1)^{1/2}(z - x_2)^{1/2}(z - x_3)^{1/2}. \qquad (11)$$

The square of dz/du is a cubic polynomial in z which Weierstrass standardized by choosing $x_1 + x_2 + x_3 = 0$ so that the cubic has no quadratic term. He wrote e_i for x_i and $\wp$ for z, whence

$$\begin{gathered}(d\wp/du)^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3) = 4\wp^3 - g_2\wp - g_3, \\ e_1 + e_2 + e_3 = 0, \qquad g_2 = -4(e_1e_2 + e_1e_3 + e_2e_3), \qquad g_3 = 4e_1e_2e_3.\end{gathered} \qquad (12)$$

This standard form was suggested by the theory of equations, in which the invariants of the general quartic polynomial, specialized to this particular cubic, are exactly g_2 and g_3.

EXAMPLE 8.2-3 (*Jacobian elliptic functions*) Much of the theory of the doubly periodic elliptic functions is guided by analogy with the singly periodic circular functions. Jacobi's elliptic functions, which include direct generalizations of the circular sine and cosine functions, antedate the more symmetrical Weierstrass function and are used more often in applied mathematics. We begin with the elliptic sine function, which reduces to the circular sine when the imaginary period becomes infinite. The corresponding reduction of the first elliptic integral to the inverse sine is

$$R_F(x, y, y) = R_C(x, y), \tag{13}$$

which follows from (6) and (6.9-14) by Theorem 5.2-4. Setting $(x_1, x_2, x_3) = (1, 0, 0)$ in (7) and using (6.9-15),

$$u(z) = R_F(z-1, z, z) = R_C(z-1, z) = \arcsin(z^{-1/2}), \qquad z^{-1/2} = \sin u. \tag{14}$$

More generally we set $(x_1, x_2, x_3) = (1, k^2, 0)$, where k is called the modulus, and define the elliptic sine function $\operatorname{sn} u = \operatorname{sn}(u, k)$ to be the resulting generalization of $\sin u$:

$$u(z) = R_F(z-1, z-k^2, z) = \operatorname{arcsn}(z^{-1/2}), \qquad z^{-1/2} = \operatorname{sn} u. \tag{15}$$

Thus sn = sin if $k^2 = 0$. If $k^2 = 1$, then sn = tanh by Exercise 6.9-16:

$$\begin{aligned} u(z) &= R_F(z-1, z-1, z) = R_C(z, z-1) \\ &= \operatorname{arctanh}(z^{-1/2}), \qquad z^{-1/2} = \tanh u. \end{aligned} \tag{16}$$

By (15),

$$u = R_F\left[\left(\frac{\operatorname{cn} u}{\operatorname{sn} u}\right)^2, \left(\frac{\operatorname{dn} u}{\operatorname{sn} u}\right)^2, \left(\frac{1}{\operatorname{sn} u}\right)^2\right], \tag{17}$$

where $\operatorname{cn} u = (1 - \operatorname{sn}^2 u)^{1/2}$ is the elliptic cosine, but $\operatorname{dn} u = (1 - k^2 \operatorname{sn}^2 u)^{1/2}$ has no circular analog.

If we write $y = z^{-1/2}$ and $u(z) = v(y)$, so that $y(v) = \operatorname{sn} v$, then (15) and Exercise 8.1-5 yield

$$\begin{aligned} v(y) &= R_F(y^{-2} - 1, y^{-2} - k^2, y^{-2}) = yR_F(1 - y^2, 1 - k^2y^2, 1) \\ &= \int_0^y (1 - t^2)^{-1/2}(1 - k^2t^2)^{-1/2}\, dt \end{aligned} \tag{18}$$

and

$$\begin{aligned} dv &= (1 - y^2)^{-1/2}(1 - k^2y^2)^{-1/2} \\ &= k^{-1}(y + 1/k)^{-1/2}(y + 1)^{-1/2}(y - 1)^{-1/2}(y - 1/k)^{-1/2}\, dy. \end{aligned} \tag{19}$$

We shall consider only the simple case $0 < k < 1$. As y describes the upper edge of a cut along the real axis in the direction of increasing real values, dy is positive and dv has constant phase except for jumps of $\frac{1}{2}\pi$ when y

passes the points $-1/k$, -1, 1, $1/k$. Hence v describes a rectangle (see Fig. 8.2-2) in the positive direction with vertices $v(-1/k)$, $v(-1)$, $v(1)$, $v(1/k)$, and the open upper half y plane is mapped conformally and one-to-one onto the interior of the rectangle. There is no vertex at $v(0) = 0$ or $v(\infty)$. By (18), $v(1) > 0$ and $v(-1) = -v(1)$. Since dv/dy is even in y, $v(\infty) - v(1/k) = v(-1/k) - v(-\infty)$. Therefore $v(\infty)$ is half way between $v(-1/k)$ and $v(1/k)$ and so is purely imaginary.

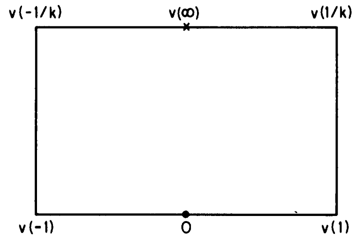

Figure 8.2-2 An image in the v plane of the upper half y plane, where $v(y) = R_F(y^{-2} - 1, y^{-2} - k^2, y^{-2}) = \operatorname{arcsn} y$ with $0 < k < 1$. The sides of the rectangle are half-periods of the inverse mapping $y = \operatorname{sn} v$, which has a pole at $v(\infty)$ and a zero at $v(0) = 0$.

Inversely $y(v) = \operatorname{sn} v$ maps the interior of the rectangle conformally and one-to-one onto the open upper half y plane. Since $y(v)$ is real on the rectangle, it can be continued analytically by successive applications of the Schwarz reflection principle to the interior of successive adjacent rectangles. Thus $\operatorname{sn} v$ is a doubly periodic function with the sides of each rectangle $[2v(1)$ and $v(\infty)]$ as half-periods. From the (simple) pole at $v(\infty)$, reflection generates a pole at $v(\infty) + 2v(1)$ and at every point displaced from either of these two by whole numbers of periods. From the (simple) zero at $v(0) = 0$, reflection generates a zero at $2v(1)$ and at every point displaced from either of these two by whole numbers of periods.

By setting $t = s^{1/2}$ in (18) and using (8.1-2), the real half-period is found to be

$$2v(1) = \int_0^1 s^{-1/2}(1 - s)^{-1/2}(1 - k^2 s)^{-1/2}\, ds = \pi R_K(1 - k^2, 1). \tag{20}$$

By Exercise 8.1-1 the imaginary half-period is

$$v(\infty) = v(1/k) - v(1) = i\int_1^{1/k} (t^2 - 1)^{-1/2}(1 - k^2 t^2)^{-1/2}\, dt = i(\pi/2)R_K(k^2, 1). \tag{21}$$

In Legendre's notation [see (6.10-10)] these half-periods are $2K$ and iK'. As $k \to 0$ the real half-period tends to π, the imaginary half-period becomes infinite [see (8.3-16)], and sn v reduces to sin v.

8.3 Gauss's theorem and dependence on a small variable

Since x^t has a branch point at $x = 0$ unless t is an integer, it is natural that $R_t(b, z)$, regarded as a function of one component of z, say z_k, has a branch point at $z_k = 0$. Just as x^t has a well-defined value at $x = 0$ if $\operatorname{Re} t > 0$, so does $R_t(b, z)$ have a well-defined value at $z_k = 0$ if t and b are restricted. To find this value we shall use the representation [see (6.8-6) or (8.1-3)]

$$\mathrm{B}(a, a')R_{-a}(b, z) = \int_0^\infty t^{a'-1} \prod_{i=1}^{k} (t + z_i)^{-b_i}\, dt, \qquad a + a' = \sum_{i=1}^{k} b_i, \tag{1}$$

which requires $a, a' \in \mathbb{C}_>$. This further restriction is inessential and will subsequently be removed by using relations between associated functions. As a corollary we shall obtain Gauss's theorem on the value of ${}_2F_1(\alpha, \beta; \gamma; 1)$, which is another notation for $R_{-\alpha}(\beta, \gamma - \beta; 0, 1)$.

THEOREM 8.3-1 Let $a, a' \in \mathbb{C}_>$, $b \in \mathbb{C}^k$, and $z \in \mathbb{C}_0{}^k$, where $k \geq 2$. Assume $a + a' = \sum_{i=1}^k b_i$ and $a' - b_k \in \mathbb{C}_>$. Then, as $z_k \to 0$ in the sector $|\mathrm{ph}\, z_k| \leq \pi - \delta < \pi$,

$$\lim \mathrm{B}(a, a')R_{-a}(b, z) = \mathrm{B}(a, a' - b_k)R_{-a}(b_1, \ldots, b_{k-1}; z_1, \ldots, z_{k-1}). \tag{2}$$

Remark By (6.3-3) we may say that $\mathrm{B}(a, a')R_{-a}(b, z)$ has the same value at $z_k = 0$ as at $b_k = 0$, since $a' - b_k = b_1 + \cdots + b_{k-1} + 0 - a$.

Proof Let S_δ be the closed set obtained by cutting a sector of angular half-width δ centered about the negative real axis out of the unit disk:

$$S_\delta = \{x \in \mathbb{C} : 0 < |x| \leq 1, |\mathrm{ph}\, x| \leq \pi - \delta < \pi\} \cup \{0\}. \tag{3}$$

If $t \geq 0$, the distance from the point $-t$ on the negative real axis to the set S_δ is at least $t \sin \delta$. Hence $z_k \in S_\delta$ implies

$$t \sin \delta \leq |t + z_k| \leq t + |z_k| \leq t + 1.$$

If we define $\theta = \mathrm{ph}(t + z_k)$, then

$$|(t + z_k)^{-b_k}| = |t + z_k|^{-\operatorname{Re} b_k} \exp(\theta \operatorname{Im} b_k).$$

Since $|\theta| < \pi$, the exponential is bounded by a positive constant A independent of z_k and t. Hence

$$|(t + z_k)^{-b_k}| \leq h(t),$$

where

$$h(t) = A(t \sin \delta)^{-\operatorname{Re} b_k}, \qquad \operatorname{Re} b_k \geq 0,$$
$$h(t) = A(t + 1)^{-\operatorname{Re} b_k}, \qquad \operatorname{Re} b_k \leq 0.$$

For every $z_k \in S_\delta$ the integrand of (1) is majorized by

$$g(t) = t^{\operatorname{Re} a' - 1} \left| \prod_{i=1}^{k-1} (t + z_i)^{-b_i} \right| h(t),$$

and g is integrable on $\mathbb{R}_+$. By the Lebesgue theorem of dominated convergence (Appendix B.1) we can evaluate the left side of (2) by taking the limit under the integral sign in (1); this yields the right side of (2). □

Whenever $R_{-a}(b, z)$ has a limit as $z_k \to 0$ in $S_\delta - \{0\}$ for every $\delta > 0$, we denote the limit by $R_{-a}(b_1, \ldots, b_k; z_1, \ldots, z_{k-1}, 0)$. The preceding theorem shows that the limit exists if $a, a', a' - b_k \in \mathbb{C}_>$, and now we shall remove the condition $a, a' \in \mathbb{C}_>$.

THEOREM 8.3-2 Let $a \in \mathbb{C}$, $b \in \mathbb{C}^k$, and $z_1, \ldots, z_{k-1} \in \mathbb{C}_0$, where $k \geq 2$. Define $c = \sum_{i=1}^k b_i$, $a' = c - a$, and $T = \{(a, b) \in \mathbb{C}^{k+1} : a' - b_k \in \mathbb{C}_>\}$. Then $R_{-a}(b_1, \ldots, b_k; z_1, \ldots, z_{k-1}, 0)/\Gamma(c)$ is holomorphic in (a, b) on T, where it has the value

$$\frac{R_{-a}(b_1, \ldots, b_k; z_1, \ldots, z_{k-1}, 0)}{\Gamma(c)} = \frac{\Gamma(a' - b_k)}{\Gamma(a')} \frac{R_{-a}(b_1, \ldots, b_{k-1}; z_1, \ldots, z_{k-1})}{\Gamma(c - b_k)}. \tag{4}$$

Remark The quantities $c - b_k = \sum_{i=1}^{k-1} b_i$ and $a' - b_k = c - b_k - a$ are independent of b_k. Also, since a function is holomorphic in several variables jointly if it is holomorphic in each one separately, both sides of (4) are holomorphic in $(a, b, z_1, \ldots, z_{k-1})$ on $T \times \mathbb{C}_0^{k-1}$.

Proof For every $n \in \mathbb{N}$ define $T_n = \{(a, b) \in T : a + n,\ a' + n \in \mathbb{C}_>\}$. Dividing (2) by $\Gamma(a)\Gamma(a')$ shows that (4) holds on T_0. Since the right side of (4) is holomorphic in (a, b) on T by Theorem 6.8-2, the left side must be holomorphic on T_0. It suffices to show that the left side is holomorphic on T, for the permanence of functional relations will then ensure that (4) holds on T.

Set $i = 1$ in (5.9-7) and apply (5.9-6) to the first term on the right side and (5.9-5) to the second term. Dividing by $\Gamma(c + 2)$ and setting $t = -a$, we find

$$\frac{R_{-a}(b, z)}{\Gamma(c)} = \sum_{i=1}^{k} (b + e_1)_i (a z_1 + a' z_i) \frac{R_{-a-1}(b + e_1 + e_i, z)}{\Gamma(c + 2)}. \tag{5}$$

Since $a' = \sum_{i=1}^k b_i - a$, changing (a, b) into $(a + 1, b + e_1 + e_i)$ changes $a, a', a' - b_k$ into $a + 1, a' + 1, a' + 1 - b_k - \delta_{ik}$. Thus $(a, b) \in T_{n+1}$ implies

$(a + 1, b + e_1 + e_i) \in T_n$ for $i = 1, \ldots, k$. If $R_{-a}(b, z)/\Gamma(c)$ has a limit as $z_k \to 0$ that is holomorphic in (a, b) on T_n, it follows from (5) that the same is true on T_{n+1}. Since it is true on T_0, it is true by induction on T_n for every $n \in \mathbb{N}$ and therefore on T. ▯

COROLLARY 8.3-3 Let $\alpha, \beta, \gamma \in \mathbb{C}$ and assume $\operatorname{Re}(\gamma - \alpha - \beta) > 0$. As $x \to 1$ in the sector $|\operatorname{ph}(1 - x)| \leq \pi - \delta < \pi$,

$$\lim \frac{{}_2F_1(\alpha, \beta; \gamma; x)}{\Gamma(\gamma)} = \frac{\Gamma(\gamma - \alpha - \beta)}{\Gamma(\gamma - \alpha)\Gamma(\gamma - \beta)}. \tag{6}$$

Remark Note that x may approach 1 from inside or outside the unit circle. The left side of (6) is usually denoted by ${}_2F_1(\alpha, \beta; \gamma; 1)/\Gamma(\gamma)$ and is holomorphic in α, β, γ if $\operatorname{Re}(\gamma - \alpha - \beta) > 0$. If $\operatorname{Re}(\gamma - \alpha - \beta) \leq 0$, the limit does not exist unless the ${}_2F_1$ function is a polynomial (Exercise 2.4-2).

Proof From (5.9-11), the symmetry of R, and the permanence of functional relations, it follows that

$${}_2F_1(\alpha, \beta; \gamma; x) = R_{-\alpha}(\gamma - \beta, \beta; 1, 1 - x). \tag{7}$$

In (4) set $k = 2$, $a = \alpha$, $b = (\gamma - \beta, \beta)$, and $z_1 = 1$. ▯

COROLLARY 8.3-4 (*Gauss's theorem*) Let $\alpha, \beta \in \mathbb{C}$ and $\gamma \in U$, and assume either $\operatorname{Re}(\gamma - \alpha - \beta) > 0$ or $-\alpha \in \mathbb{N}$ or $-\beta \in \mathbb{N}$. Then

$${}_2F_1(\alpha, \beta; \gamma; 1) = \sum_{n=0}^{\infty} \frac{(\alpha, n)(\beta, n)}{(\gamma, n)n!} = \frac{\Gamma(\gamma)\Gamma(\gamma - \alpha - \beta)}{\Gamma(\gamma - \alpha)\Gamma(\gamma - \beta)}. \tag{8}$$

Proof We need to consider only the case $\operatorname{Re}(\gamma - \alpha - \beta) > 0$ since the other cases follow from Exercise 2.4-2. If we define $u_n = (\alpha, n)(\beta, n)/(\gamma, n)n!$, then by (3.4-5),

$$u_n \sim C n^{\alpha + \beta - \gamma - 1}, \qquad n \to \infty,$$

where C is a constant. Since $\sum n^{-s-1}$ converges if $s > 0$, the series in (8) converges absolutely by the limit comparison test [Apostol (1974, Section 8.10)]. Therefore, $\sum u_n x^n$ converges uniformly for $|x| \leq 1$, its sum ${}_2F_1(\alpha, \beta; \gamma; x)$ is continuous for $|x| \leq 1$, and the limit of the sum as $x \to 1$ is

$${}_2F_1(\alpha, \beta; \gamma; 1) = \sum_{n=0}^{\infty} u_n.$$

The equality of the first and third members of (8) follows from (6). ▯

Whether or not $R_t(b, z)$ has a well-defined value at $z_k = 0$, it is sometimes useful to know its behavior for small $|z_k|$. We shall give only a brief sketch. Consider the $\binom{k+2}{2}$ distinct integrals [cf. Exercise 5.4-7],

$$u_{rs} = \int_{-z_r}^{-z_s} t^{a' - 1} \prod_{i=1}^{k} (t + z_i)^{-b_i}\, dt = -u_{sr}, \qquad r < s, \tag{9}$$

where r and s take the values $0, 1, \ldots, k, \infty$, with $z_0 = 0$ and $-z_\infty = \infty$. In particular $u_{0\infty}$ is the right side of (1). The path of integration lies in the t plane cut from each of the points $0, -z_1, \ldots, -z_k$ along a half-line parallel to the negative real axis. For convergence it is assumed that a, a', $1 - b_1, \ldots, 1 - b_k \in \mathbb{C}_>$, but this assumption can later be removed by the permanence of functional relations. By Cauchy's integral theorem one can establish linear relations (for example $u_{rs} = u_{rp} + u_{ps}$) which express all the u's in terms of $u_{1\infty}, \ldots, u_{k\infty}$. Hence there is a linear relation between any $k + 1$ integrals.

If $k = 2$, for example, there is a linear relation between $u_{0\infty}$, $u_{1\infty}$, and u_{02}. These three integrals are expressible by (8.1-3), (8.1-2), and (5.9-19) as constant multiples of the functions

$$R_{-a}(b_1, b_2; z_1, z_2), \qquad R_{-a}(1 - a', b_2; z_1, z_1 - z_2)$$
$$z_2^{b_1 - a} R_{-b_1}(1 - b_2, a'; z_1, z_1 - z_2).$$

The coefficients of the linear relation can be determined with the help of (2) by considering the case $z_1 = z_2$, $\mathrm{Re}(a + a') < 1$, and the case $z_2 = 0$, $\mathrm{Re}(b_1 - a) > 0$. The resulting connection formula, in slightly different notation, is the following. Let x, y, $x - y \in \mathbb{C}_0$ and $\alpha, \alpha', \beta, \beta' \in \mathbb{C}$. Assume $\alpha + \alpha' = \beta + \beta'$ and $\alpha - \beta \notin \mathbb{Z}$. Then

$$\frac{R_{-\alpha}(\beta, \beta'; x, y)}{\Gamma(\beta + \beta')} = \frac{\Gamma(\beta - \alpha)}{\Gamma(\alpha')\Gamma(\beta)} R_{-\alpha}(1 - \alpha', \beta'; x, x - y) + \frac{\Gamma(\alpha - \beta)}{\Gamma(\alpha)\Gamma(\beta')} y^{\beta - \alpha} R_{-\beta}(1 - \beta', \alpha'; x, x - y). \tag{10}$$

If $\alpha' - \beta' \in \mathbb{C}_>$ (whence $\beta - \alpha \in \mathbb{C}_>$) the second term on the right tends to 0 as $y \to 0$, and (10) agrees with (4). If $n \in \mathbb{N}$ and $\alpha = -n$, the second term on the right vanishes and (10) reduces to (6.5-1).

If $\alpha - \beta \in \mathbb{Z}$, the right side of (10) is indeterminate and must be evaluated by taking a limit. An illustrative example of special interest is the complete elliptic integral R_K. If $|x| > |y|$, $\beta = \beta' = \frac{1}{2}$, $\alpha = \frac{1}{2} + \varepsilon$, $\alpha' = \frac{1}{2} - \varepsilon$, we use (5.9-23) and (3.9-3) to obtain

$$R_{-1/2-\varepsilon}\left(\frac{1}{2}, \frac{1}{2}; x, y\right) = (\pi x)^{-1/2} \cot(\pi\varepsilon)$$
$$\cdot \left[-x^{-\varepsilon} \sum_{n=0}^{\infty} \frac{\Gamma(\frac{1}{2} + \varepsilon + n)(\frac{1}{2}, n)}{\Gamma(1 + \varepsilon + n) n!} \left(\frac{y}{x}\right)^n + y^{-\varepsilon} \sum_{n=0}^{\infty} \frac{\Gamma(\frac{1}{2} - \varepsilon + n)(\frac{1}{2}, n)}{\Gamma(1 - \varepsilon + n) n!} \left(\frac{y}{x}\right)^n \right]. \tag{11}$$

As $\varepsilon \to 0$,

$$\cot(\pi\varepsilon) \sim 1/\pi\varepsilon, \qquad x^{-\varepsilon} = e^{-\varepsilon \log x} \sim 1 - \varepsilon \log x,$$

$$\frac{\Gamma(\frac{1}{2} + \varepsilon + n)}{\Gamma(1 + \varepsilon + n)} \sim \frac{\Gamma(\frac{1}{2} + n) + \varepsilon\Gamma'(\frac{1}{2} + n)}{\Gamma(1 + n) + \varepsilon\Gamma'(1 + n)} \sim \pi^{1/2} \frac{(\frac{1}{2}, n)}{(1, n)} \cdot \left[1 + \varepsilon\psi\left(\frac{1}{2} + n\right) - \varepsilon\psi(1 + n)\right], \tag{12}$$

where the psi function is defined by (3.6-12). Therefore,

$$R_K(x, y) = \frac{x^{-1/2}}{\pi} \sum_{n=0}^{\infty} \frac{(\frac{1}{2}, n)(\frac{1}{2}, n)}{(1, n)n!} \left[\log(x/y) + 2\psi(1 + n) - 2\psi\left(\frac{1}{2} + n\right)\right]\left(\frac{y}{x}\right)^n. \tag{13}$$

By (5.9-23),

$$\pi R_K(x, y) = \log(x/y) R_K(x, x - y) + 2x^{-1/2} \sum_{n=0}^{\infty} \frac{(\frac{1}{2}, n)^2}{(n!)^2} \cdot \left[\psi(1 + n) - \psi\left(\frac{1}{2} + n\right)\right]\left(\frac{y}{x}\right)^n. \tag{14}$$

The quantity in brackets can be evaluated by (3.6-10) and Exercises 3.7-5 and 3.6-5:

$$\psi(1) - \psi(\tfrac{1}{2}) = 2 \log 2, \tag{15}$$

$$\psi(1 + n) - \psi\left(\frac{1}{2} + n\right) = 2\left(\log 2 - 1 + \frac{1}{2} - \frac{1}{3} + \cdots - \frac{1}{2n - 1} + \frac{1}{2n}\right), \quad n \geq 1.$$

By (14), $R_K(x, y)$ has a logarithmic singularity at $y = 0$. If $x, y \in \mathbb{C}_0$ and $y \to 0$ (or $x \to \infty$ with y fixed because y/x is what matters), then

$$R_K(x, y) \sim (x^{-1/2}/\pi)\log(16x/y). \tag{16}$$

This is a special case (see Exercise 9.2-1) of an asymptotic formula (9.2-10) for the incomplete elliptic integral R_F defined by (8.2-6). The connection between the complete and incomplete integrals is a special case of (2),

$$R_F(x, y, 0) = (\pi/2)R_K(x, y). \tag{17}$$

8.4 Existence theorem for associated functions

If t and b are fixed while $n, m_1, \ldots, m_k$ range over the integers, all functions of the form $R_{t+n}(b+m, z)$ are associated R functions according to Section 5.6 and (5.9-2). Relations 5.9-5 give three examples of linear homogeneous relations between associated R functions with coefficients that are polynomials in $z_1, \ldots, z_k$. We shall now prove that there exists a linear homogeneous relation with polynomial coefficients between any $k+1$ associated R functions in k variables. This theorem will have important consequences in Section 9.2 for the theory of elliptic integrals.

The first step is to prove a recurrence relation for the homogeneity parameter when all b parameters remain fixed. [See (5.9-25) for the case $k=2$ and Exercise 5.9-9 for $k=3$.] The theorem follows from this relation without much difficulty.

RELATION 8.4-1 Let $a \in \mathbb{C}$, $z \in \mathbb{C}_0{}^k$, $b \in U_k$, and $a + a' = \sum_{i=1}^k b_i$. Define the elementary symmetric function $E_n(z)$ by $(1+xz_1)\cdots(1+xz_k) = \sum_{n=0}^k x^n E_n(z)$, and let $D_i = \partial/\partial z_i$. Then

$$\sum_{n=0}^{k} A_n(a, b, z) R_{-a-n}(b, z) = 0, \tag{1}$$

$$A_n(a, b, z) = \frac{(a, n)(a'-k, k-n)}{a(a'-k)} \left(a + n - \sum_{i=1}^{k} b_i z_i D_i \right) E_n(z). \tag{2}$$

Remark Despite the denominator in (2), A_n is a polynomial in a, b, z. This is true plainly for $A_1, \ldots, A_{k-1}$ and also for $A_0 = (a'-k+1, k-1)$ and $A_k = -(a+1, k-1)z_1 \cdots z_k$.

Proof Since A_n is a polynomial, Theorem 6.8-2 ensures that the left side of (1) is holomorphic in a, b, z. By the permanence of functional relations it suffices to prove (1) if $a, a'-k \in \mathbb{C}_>$. By Exercise 6.8-8,

$$\mathrm{B}(a, a') R_{-a}(b, z) = \int_0^\infty x^{a-1} \prod_{i=1}^{k} (1+xz_i)^{-b_i}\, dx, \qquad a, a' \in \mathbb{C}_>. \tag{3}$$

For brevity let

$$P = \prod_{i=1}^{k} (1+xz_i)^{-b_i}, \qquad G = \prod_{i=1}^{k} (1+xz_i) = \sum_{n=0}^{k} x^n E_n, \tag{4}$$

and differentiate the last equation with respect to z_i to obtain

$$x(1+xz_i)^{-1} G = \sum_{n=0}^{k} x^n D_i E_n. \tag{5}$$

Now consider

$$\frac{d}{dx}\left[x^a \prod_{i=1}^{k}(1 + xz_i)^{1-b_i}\right] = ax^{a-1}GP + x^a \sum_{i=1}^{k}(1 - b_i)z_i(1 + xz_i)^{-1}GP$$

$$= x^{a-1}P\left[a\sum_{n=0}^{k} x^n E_n + \sum_{i=1}^{k}(1 - b_i)z_i \sum_{n=0}^{k} x^n D_i E_n\right]$$

$$= \sum_{n=0}^{k} x^{a+n-1}P\left[aE_n + \sum_{i=1}^{k}(1 - b_i)z_i D_i E_n\right]. \tag{6}$$

Since E_n is homogeneous of degree n, $\sum_{i=1}^{k} z_i D_i E_n = nE_n$. Integrating (6) with respect to x from 0 to ∞ and using (3), we obtain

$$0 = \sum_{n=0}^{k} \mathrm{B}(a + n, a' - n)R_{-a-n}(b, z)\left(a + n - \sum_{i=1}^{k} b_i z_i D_i\right)E_n. \tag{7}$$

Multiplication by $\Gamma(a + a')/\Gamma(a + 1)\Gamma(a' - k + 1)$ yields (1). □

LEMMA 8.4-2 Let $n \in \mathbb{Z}$, $z \in \mathbb{C}_0{}^k$, $b \in U_k$, and $c = \sum_{i=1}^{k} b_i$. Let $1 - t \in U$ and assume $c + t - k + 2 \in U$. Then

$$R_{t+n}(b, z) = \sum_{p=1}^{k} \theta_p(z)R_{t+1-p}(b, z), \tag{8}$$

where $\theta_p(z)$ is a rational function of $z_1, \ldots, z_k, b_1, \ldots, b_k, t, n, p$.

Remark This shows that the k functions $R_t, R_{t-1}, \ldots, R_{t-k+1}$, although not proved to be linearly independent, at least contain a basis for all associated R functions with the same b parameters. The condition $1 - t \in U$ does not allow t to be a positive integer, for in that case two or more of the chosen functions would be polynomials and so could not be linearly independent with respect to rational coefficients. Similarly, if $k - c - t - 1$ were a positive integer, two or more functions would be rational functions times $\prod z_i^{-b_i}$ by (6.8-15).

Proof If $n = 0, -1, \ldots, -k + 1$, (8) is trivial. If $n = -k$, set $-a = t$ in (1) and divide by A_k to obtain (8). The division is permissible because the condition $a + 1 = 1 - t \in U$ ensures that $A_k = -(a + 1, k - 1)z_1 \cdots z_k$ is not the zero polynomial. Hence R_{t-k} is a linear combination of $R_{t-k+1}, \ldots, R_t$ with rational coefficients. If $n = -k - 1$, set $-a = t - 1$ in (1), note that $a + 1 = 2 - t \in U$, and divide by A_k to show that R_{t-k-1} is a linear combination of $R_{t-k}, \ldots, R_{t-1}$ and hence of $R_{t-k+1}, \ldots, R_t$. Repetition of the procedure proves inductively that (8) holds for every negative integer n.

If $n = 1$, set $-a = t + 1$ in (1) and divide by $A_0 = (a' - k + 1, k - 1)$ which is permissible because $a' - k + 1 = c - a - k + 1 = c + t - k + 2 \in U$. Thus R_{t+1} is a linear combination of $R_t, \ldots, R_{t-k+1}$. If $n = 2$, set $-a = t + 2$ in (1), note that $a' - k + 1 = c + t - k + 3 \in U$, and divide by A_0 to show that R_{t+2} is a linear combination of $R_{t+1}, \ldots, R_{t-k+2}$ and hence of $R_t, \ldots, R_{t-k+1}$. Repetition of the process proves inductively that (8) holds for every positive integer n. ▯

THEOREM 8.4-3 Between any $k + 1$ associated R functions of k variables there exists a linear homogeneous relation with polynomial coefficients.

Proof The $k + 1$ associated functions necessarily have the form $R_{t+n}(b + m, z)$, where t and b are fixed while $n, m_1, \ldots, m_k$ have integral values that differ from one function to another. If we call a k-tuple a vector, we have to consider $k + 1$ vectors m, one for each function. Clearly we can choose a single vector B such that all components of $B - b - m$ are nonnegative integers for every one of the $k + 1$ vectors m. (In case $\sum b_i$ is an integer, we further require $\sum B_i > k - 2$.) By successive applications of (5.9-7), which adds unity to a single b parameter, we can express each of the $k + 1$ associated functions as a finite linear combination,

$$R_{t+n}(b + m, z) = \sum_N \alpha_N(z) R_{t+N}(B, z). \tag{9}$$

The index N takes integral values and the coefficient $\alpha_N(z)$ is a polynomial. (See Exercise 8.4-2 for explicit coefficients.)

We now choose T so that $T - t$ is an integer, $1 - T \in U$, and $T - k + 2 + \sum_{i=1}^k B_i \in U$. (Even if $\sum B_i$ is an integer, these requirements are compatible because we chose $\sum B_i > k - 2$.) By Lemma 8.4-2 each term on the right side of (9) can be written as a linear combination of $R_T(B, z), \ldots, R_{T+1-k}(B, z)$ with coefficients that are rational functions. Thus each of the original $k + 1$ associated functions is a linear combination,

$$R_{t+n}(b + m, z) = \sum_{p=1}^{k} \phi_p(z) R_{t+1-p}(B, z). \tag{10}$$

where $\varphi_p(z)$ is a rational function of z which depends on n and m. Between the $k + 1$ equations of this form we can eliminate the k functions $R_T(B, z), \ldots, R_{T+1-k}(B, z)$ to obtain a linear homogeneous relation between the original $k + 1$ associated functions. The coefficients will be rational functions of z, and multiplication by the lowest common denominator of the rational functions will change the coefficients into polynomials. ▯

8.5 Removal of integral parameters

It may be possible to express each member of a large class of R functions (not necessarily all associated) in terms of elementary functions and a small number of chosen R functions. Reduction to these standard functions results in part from linear dependence of associated functions (Section 8.4) and in part from removal of parameters with integral values. Removal of parameters will be illustrated first by the case in which all parameters are integers.

THEOREM 8.5-1 Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^k \cap U_k$. Then $R_{-a}(b, z)$ is expressible in terms of rational functions and logarithms.

Proof Any b parameter that is zero can be eliminated by (6.3-3). Any b parameter that is a negative integer can be raised by successive applications of (5.9-7) until it reaches zero and is eliminated. Thus we may assume that b is a k-tuple of positive integers. If $-a \in \mathbb{N}$, then R_{-a} is a polynomial, and if $-a' \in \mathbb{N}$, where $a + a' = c = \sum_{i=1}^k b_i$, then R_{-a} is a rational function by (6.8-15). Hence we suppose that a and a' also are positive integers.

By (5.6-10) with $F = R_{1-a}$ (or by Exercise 5.9-6),

$$(a-1)(z_i - z_j)R_{-a}(b, z) = (c-1)[R_{1-a}(b - e_i, z) - R_{1-a}(b - e_j, z)]. \tag{1}$$

This holds for all $i, j = 1, \ldots, k$ and (by the permanence of functional relations) for $a \in \mathbb{C}$, $b \in U_k$, $z \in \mathbb{C}_0{}^k$. Provided $a > 1$, $R_{-a}(b, z)$ is expressed by (1) as a combination of functions in which one b parameter is lowered by unity and thus eliminated if it was unity initially. We apply (1) repeatedly, changing the choice of i or j whenever b_i or b_j has been lowered to zero and eliminated. Since a also is lowered by unity each time, successive applications must stop when a reaches the value 1 because of the factor $a - 1$ on the left side. However, since a' is unaffected by (1), interchanging a and a' by (6.8-15) allows the successive applications to resume and continue until $a = a' = 1$. Since $a + a' = 2 = \sum_{i=1}^k b_i$, the original function has now been expressed as a finite linear combination (with coefficients that are rational in $z_1, \ldots, z_k$) of the functions $R_{-1}(2, 0; z_i, z_j) = z_i^{-1}$ and

$$R_{-1}(1, 1; z_i, z_j) = \frac{\log z_i - \log z_j}{z_i - z_j}, \qquad i, j = 1, \ldots, k. \tag{2}$$

(See Exercise 5.9-13.) ▯

Even if not all the parameters are integers, some of those that are integers can be removed by this method. The number of nonintegral parameters remains the same and is therefore more significant than the total number of parameters.

DEFINITION 8.5-2 Assume $z \in \mathbb{C}_0{}^k$, $a \in \mathbb{C}$, $b \in U_k$, and $a + a' = \sum_{i=1}^k b_i$. Then $R_{-a}(b, z)$ is said to have index h if exactly h of the parameters $a, a', b_1, \ldots, b_k$ are not integers.

Remark The index h cannot be unity because of the relation between the parameters. Since Theorem 8.5-1 disposes of the case $h = 0$, it suffices to consider $h \geq 2$.

In removing integers it is helpful first to effect a standard arrangement of parameters in which a' is an integer and a is not. This has the advantage that the coefficient $a - 1$ on the left side of (1) cannot vanish, but a more compelling reason is convenience in the important cases considered at the end of this section and in Chapter 9.

THEOREM 8.5-3 Let $R_{-a}(b, z)$ be a function of k variables with index $h \geq 2$. Then it is expressible in terms of rational functions, powers (with exponents $-b_1, \ldots, -b_k$), and functions of h variables of the form $R_{-\alpha}(\beta, w)$, where α' is a positive integer and β_h a nonnegative integer (α' being defined by $\alpha + \alpha' = \sum_{i=1}^h \beta_i$) while $\alpha, \beta_1, \ldots, \beta_{h-1}$ are not integers. The latter parameters differ by integers from the elements of a subset (possibly different for each function) of $\{a, a', b_1, \ldots, b_k, -a, -a', -b_1, \ldots, -b_k\}$.

Proof There are four possibilities:

(i) a' is an integer but a is not,
(ii) a is an integer but a' is not,
(iii) neither a nor a' is an integer,
(iv) both a and a' are integers.

We shall show that the last three cases can be reduced to the first. Case (ii) is disposed of at once by using (6.8-15) to interchange a and a'.

In case (iii) we may suppose $a + a' \in \mathbb{C}_>$ because unity can be added to $a + a' = \sum_{i=1}^k b_i$ as many times as necessary by successive applications of (5.9-7). Then at least one of a and a' must be in $\mathbb{C}_>$, and by (6.8-15) we may assume that it is a'. By (8.3-4) we may insert an additional b parameter and a corresponding variable with value 0:

$$\begin{aligned} &\mathrm{B}(a, a')R_{-a}(b_1, \ldots, b_k; z_1, \ldots, z_k) \\ &\qquad = \mathrm{B}(a, 1)R_{-a}(b_1, \ldots, b_k, 1 - a'; z_1, \ldots, z_k, 0). \end{aligned}$$

The function on the right belongs to case (i).

In case (iv) the R function is a polynomial if $-a \in \mathbb{N}$ or, by (6.8-15), a polynomial times a product of powers if $-a' \in \mathbb{N}$. Therefore we assume that a and a' are positive integers, and we may then use (6.8-6),

$$B(a, a')R_{-a}(b, z) = \int_0^\infty t^{a'-1} \prod_{i=1}^{k} (t + z_i)^{-b_i}\, dt. \tag{3}$$

If a or a' were not an integer, one limit of integration would be a branch point of the integrand. Conversely, replacing one limit of integration by a branch point ($-z_i$ for a suitable value of i) should yield an R function belonging to case (i) or (ii). To this end we permute the b parameters and variables so that $b_1, \ldots, b_h$ are not integers and none of the points $-z_2, \ldots, -z_h$ lies in the closed convex hull of the point $-z_1$ and the positive real axis. (This is possible because $z \in \mathbb{C}_0{}^k$ and Theorem 5.2-4 allows us to suppose that no two components of z are equal.) By Cauchy's integral theorem we may then split the integral into two parts,

$$\int_0^\infty = \int_{-z_1}^\infty - \int_{-z_1}^0, \tag{4}$$

assuming for the moment that $\operatorname{Re} b_1 < 1$ to ensure convergence. The first integral on the right can be evaluated by (8.1-3) and then transformed by (6.8-15); the second integral can be evaluated by (8.1-2). The result is

$$\begin{aligned} B(a, a')R_{-a}(b, z) &= (-1)^{a'-1} \prod_{i=2}^{k} (z_i - z_1)^{-b_i} \\ &\cdot \Big\{ B(1 - b_1, a) z_1^{a'-1} R_{b_1-1}\Big(1 - a', b_2, \ldots, b_k; \frac{-1}{z_1}, \frac{1}{z_2 - z_1}, \ldots, \frac{1}{z_k - z_1}\Big) \\ &- B(1 - b_1, a') z_1^{a'-b_1} R_{b_1-1}\Big(1 - a, b_2, \ldots, b_k; 1, \frac{z_2}{z_2 - z_1}, \ldots, \frac{z_k}{z_k - z_1}\Big)\Big\}. \end{aligned} \tag{5}$$

[Even if $-1/z_1$ is real and negative, the first R function on the right side is well-defined in virtue of (5.9-7) because the corresponding parameter $1 - a'$ is a nonpositive integer.] By Theorem 6.8-2 and the permanence of functional relations, (5) is valid if $1 - b_1 \in U$. Both R functions on the right belong to case (i).

All cases have now been reduced to case (i), and by permutation symmetry we need to consider only functions of m variables, $m \le k + 1$, of the form $R_{-\alpha}(\beta, w)$, where α', $\beta_h, \ldots, \beta_m$ are integers and α, $\beta_1, \ldots, \beta_{h-1}$ are not integers. In case (iii) we introduced $1 - a'$ as a nonintegral β parameter, and in case (iv) we chose $\alpha = 1 - b_1$, where b_1 stood for some nonintegral b parameter.

If α' is a nonpositive integer, $R_{-\alpha}(\beta, w)$ is a product of powers times a polynomial by (6.8-15). If α' is a positive integer, any b parameter that is a negative integer will be raised to 0 by successive applications of (5.9-7) and eliminated (unless $m = h$) by (6.3-3). In the only case requiring further discussion, $\beta_h, \ldots, \beta_m$ are positive integers and $m > h$. By successive applications of (1), with i and j chosen from among $h, \ldots, m$, we can lower $\beta_h, \ldots, \beta_m$ until all but one reach 0 and are eliminated. When only one is left, the process stops because the left side of (1) vanishes if $i = j$. ▯

This theorem has consequences of great practical importance when every one of the parameters a, a', $b_1, \ldots, b_k$ is either an integer or half of an odd integer. The parameters α, $\beta_1, \ldots, \beta_{h-1}$ must be half-odd-integers, and the condition $a + a' = \sum_{i=1}^{k} b_i$ implies that h must be even. The case $h = 2$ is the subject of the next theorem, and the case $h = 4$ will be discussed at length in the next chapter.

THEOREM 8.5-4 Assume that exactly two of the parameters $a, a', b_1, \ldots, b_k$ are half-odd-integers and the rest are integers (a' being defined by $a + a' = \sum_{i=1}^{k} b_i$). Then $R_{-a}(b, z)$ can be expressed in terms of rational functions, square roots, and logarithms.

Remark In the complex domain there is no significant distinction between logarithms, inverse circular functions, and inverse hyperbolic functions, all of them being equivalent to $R_C(x, y) = R_{-1/2}(\frac{1}{2}, 1; x, y)$ (see Example 6.9-4).

Proof By Theorem 8.5-3, $R_{-a}(b, z)$ is expressible in terms of rational functions, square roots, and functions of two variables of the form $R_{-\alpha}(\beta, w)$, where α' and β_2 are integers while α and β_1 are half-odd-integers. By Theorem 8.4-3 there is a linear relation with polynomial coefficients connecting any such function with the standard functions $R_{-1/2}(\frac{1}{2}, 1; w_1, w_2) = R_C(w)$ and $R_{-1/2}(\frac{1}{2}, 0; w_1, w_2) = w_1^{-1/2}$. Since R_C involves a logarithm, the two standard functions are linearly independent with respect to polynomial coefficients. Hence the coefficient of $R_{-\alpha}(\beta, w)$ in the linear relation cannot be the zero polynomial, and we can solve for $R_{-\alpha}(\beta, w)$ in terms of rational functions, square roots, and a logarithm. ▯

The linear relations invoked in the preceding proof serve to express $R_{-a}(b_1, b_2; x, y)$, where a and b_1 are half-odd-integers and b_2 is an integer, in terms of the standard functions $R_C(x, y)$ and $R_{-1/2}(\frac{1}{2}, 0; x, y) = x^{-1/2}$. The first step in finding explicit coefficients is to raise or lower the b parameters by successive applications of (5.9-7) and (5.9-24) until (b_1, b_2)

reaches $(\frac{1}{2}, 0)$ or $(\frac{1}{2}, 1)$. In the latter case successive applications of (5.9-25) raise or lower the homogeneity parameter to the values $-\frac{1}{2}$ and $-\frac{3}{2}$. The former is the value for R_C and the latter for $R_{-3/2}(\frac{1}{2}, 1; x, y)$, which equals $x^{-1/2}y^{-1}$ by (5.9-22).

Some coefficients obtained in this way are listed in Table 8.5-1. Cases with a and b_1 both negative are unlikely to occur in practice because a simple integral representation does not exist.

Table 8.5-1

Reduction of R functions[a]

(b_1, b_2)	(a, a')	C	C_C	C_A
(1/2, 1)	(−1/2, 2)	2	y	x
(3/2, 1)	(1/2, 2)	$2(x-y)$	$-3y$	$3x$
(3/2, 1)	(3/2, 1)	$x-y$	3	-3
(−1/2, 2)	(1/2, 1)	$2y$	y	x
(1/2, 2)	(1/2, 2)	$4(x-y)$	$3(2x-y)$	$-3x$
(1/2, 2)	(3/2, 1)	$2(x-y)y$	$-3y$	$3x$
(3/2, 2)	(3/2, 2)	$4(x-y)^2$	$15(2x+y)$	$-45x$

[a] In each case $a', b_2 \in \{1, 2\}$ and $a, b_1 \in \{-\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\}$, where $a + a' = b_1 + b_2$. Cases with $a = b_1 = -\frac{1}{2}$ are omitted. The listed coefficients are those in

$$CR_{-a}(b_1, b_2; x, y) = C_C R_C(x, y) + C_A x^{-1/2}.$$

EXAMPLE 8.5-5 (*Evaluation of elementary integrals*) Together with the three integrals in Section 8.1, the table of reduction formulas (which can be extended to other values of the parameters) provides a compact table of integrals. For example, by (8.1-2) and the next to the last line of Table 8.5-1, we can evaluate the following integral in which $\lambda \neq 0$ and $\beta + \lambda t > 0$ for $\alpha \leq t < x$:

$$\begin{aligned}
&\int_\alpha^x (t-\alpha)^{1/2}(\beta + \lambda t)^{-1/2}\,dt \\
&\quad = \mathrm{B}\left(\frac{3}{2}, 1\right)(x-\alpha)^{3/2}(\beta+\lambda\alpha)^{-1/2} R_{-3/2}\left(\frac{1}{2}, 2; \frac{\beta+\lambda x}{\beta+\lambda\alpha}, 1\right) \\
&\quad = \frac{2}{3}(x-\alpha)^{3/2}(\beta+\lambda\alpha) R_{-3/2}\left(\frac{1}{2}, 2; \beta+\lambda x, \beta+\lambda\alpha\right) \\
&\quad = \lambda^{-1}(x-\alpha)^{1/2}[(\beta+\lambda x)^{1/2} - (\beta+\lambda\alpha)R_C(\beta+\lambda x, \beta+\lambda\alpha)]. \qquad (6)
\end{aligned}$$

We can express R_C in terms of a logarithm by (6.9-16) if $\lambda > 0$ or an arccosine by (6.9-15) if $\lambda < 0$. Although these cases must be distinguished in conventional integral tables, the distinction is eliminated if we compute R_C by Borchardt's algorithm (Example 6.10-5).

Many integrals with trigonometric or hyperbolic integrands can be evaluated after a change of integration variable to put the integral in the form (8.1-1). Elliptic integrals can be evaluated similarly by using the tables in Section 9.3.

NOTES

The Schwarz–Christoffel mapping is written in terms of Lauricella's function F_D by Haeusler (1966); although F_D is equivalent to R [see Carlson (1963)], the simplicity of (8.2-4) comes from choosing ∞ rather than 0 as the reference point in (8.2-2). Applications of the Schwarz–Christoffel mapping to electromagnetism and hydrodynamics are discussed by Smythe (1968, Chapter IV), Bateman (1932, Chapter IV), Oberhettinger (1949), Bowman (1961, Chapter VII), and Henrici (1974, Volume 1, Sections 5.12–5.14); Henrici considers also polygons with rounded corners.

Copson (1935) gives a good introduction to elliptic functions, Whittaker (1927) a more detailed discussion, Tricomi (1948) a thorough exposition, and Erdélyi (1953, Chapter XIII) a summary (with detailed conformal mappings in Section 13.25). See also Siegel (1969) and Rauch (1973). Jacobian functions (but not Weierstrass' function) are treated in an elementary way with applications by Bowman (1961) and in depth by Cayley (1895). Various applications are described by Tricomi (1948, Chapter V) and Oberhettinger (1949). An article by Todd (1975) on the perimeter of Bernoulli's lemniscate includes historical notes and a bibliography.

Connection formulas such as (8.3-10) are discussed (in terms of ${}_2F_1$) more fully by Lebedev (1965, Sections 9.5 and 9.7) and Erdélyi (1953, Sections 2.9 and 2.10). Carlson (1963) gives some extensions of Gauss's theorem and a proof of Theorem 8.4-3. For questions about integration related to Theorems 8.5-1 and 8.5-4 see Hardy (1916), Ritt (1948), and Rosenlicht (1972).

EXERCISES

8.1-1 Let $x, y \in \mathbb{R}$, $\lambda \in \mathbb{C}$, and $\alpha, \beta \in \mathbb{C}_>$. Assume $y > x > 0$. Show that

$$\int_x^y (t^2 - x^2)^{\alpha-1}(y^2 - t^2)^{\beta-1}t^{1-2\lambda}\,dt$$
$$= \tfrac{1}{2}\mathrm{B}(\alpha, \beta)(y^2 - x^2)^{\alpha+\beta-1}x^{2\alpha-2\lambda}R_{-\alpha}(\alpha + \beta - \lambda, \lambda; x^2, y^2).$$

8.1-2 Let $\beta, k \in \mathbb{C}$, $-\pi/2 < \varphi < \pi/2$, and $1 - k^2\sin^2\varphi \in \mathbb{C}_0$. Show that

$$\int_0^\varphi (1 - k^2\sin^2\theta)^{-\beta}\,d\theta = (\sin\varphi)R_{-1/2}\left(\frac{1}{2}, \beta, 1 - \beta; \cos^2\varphi, 1 - k^2\sin^2\varphi, 1\right),$$

where $|\mathrm{ph}(1 - k^2 \sin^2 \theta)| < \pi$. The cases $\beta = \frac{1}{2}$ and $\beta = -\frac{1}{2}$ are Legendre's incomplete elliptic integrals of the first and second kinds (see Example 9.3-1).

8.1-3 Let $a \in \mathbb{C}_>$ and $\alpha, \beta, \gamma, \delta, k \in \mathbb{C}$. Assume $0 < \varphi < \pi/2$ and $1 - k^2 \sin^2 \varphi$, $1 - \alpha^2 \sin^2 \varphi \in \mathbb{C}_0$. Show that

$$\int_0^\varphi (\sin\theta)^{2a-1}(\cos\theta)^{1-2\beta}(1 - k^2 \sin^2\theta)^{-\gamma}(1 - \alpha^2 \sin^2\theta)^{-\delta}\, d\theta$$

$$= \frac{1}{2a}(\sin\varphi)^{2a} R_{-a}(\beta, \gamma, 1 + a - \beta - \gamma - \delta, \delta; \cos^2\varphi, 1 - k^2\sin^2\varphi, 1, 1 - \alpha^2\sin^2\varphi),$$

where $|\mathrm{ph}(1 - k^2 \sin^2\theta)| < \pi$ and $|\mathrm{ph}(1 - \alpha^2 \sin^2\theta)| < \pi$. The case $a = \beta = \gamma = \frac{1}{2}$ and $\delta = 1$ is Legendre's incomplete elliptic integral of the third kind (see Exercise 9.3-1).

8.1-4 Let $p, q \in \mathbb{C} - \mathbb{R}$. Define $d = |p - q|$, $\delta = |p - \bar{q}|$, $d_> = \max\{d, \delta\}$, $d_< = \min\{d, \delta\}$. Show that

$$\int_{-\infty}^{\infty} |t - p|^{-1}|t - q|^{-1}\, dt = 2\pi R_K(d_>^2, d_>^2 - d_<^2) = 2\pi R_K[(d + \delta)^2, (d - \delta)^2],$$

where R_K is the elliptic integral defined by (6.10-4). (Cf. Exercise 9.8-4.)

8.1-5 Let $\alpha \in \mathbb{C}_>$ and $x, \beta, \delta, A, B, C, D \in \mathbb{C}$. Assume $(A + Bt^2)(C + Dt^2) \neq 0$ for every $t \in [0, x]$. Show that

$$\int_0^x t^{2\alpha-1}(A + Bt^2)^\beta(C + Dt^2)^\delta\, dt$$

$$= \frac{x^{2\alpha}}{2\alpha} A^\beta C^\delta R_{-\alpha}\left(-\beta, -\delta, 1 + \alpha + \beta + \delta; \frac{A + Bx^2}{A}, \frac{C + Dx^2}{C}, 1\right).$$

8.1-6 Let $a, a' \in \mathbb{C}$ and $b, x, y \in \mathbb{C}_>^k$. Assume $a + a' = \sum_{i=1}^k b_i$, define μ_b by (4.4-1), and let x/y denote the k-tuple with ith component x_i/y_i. Show that

$$\int_E (u \cdot x)^{-a}(u \cdot y)^{-a'}\, d\mu_b(u) = \prod_{i=1}^k y_i^{-b_i} R_{-a}(b, x/y) = \prod_{i=1}^k x_i^{-b_i} R_{-a'}(b, y/x).$$

8.2-1 Let $x \in \mathbb{R}^4$ and assume that the components of x are distinct. Find the image of the upper half z plane under the mapping

$$w(z) = R_{-1}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; z - x_1, z - x_2, z - x_3, z - x_4).$$

8.2-2 Let $x \in \mathbb{R}^k$, $k > 2$, and assume that the components of x are distinct. Find the image of the upper half z plane under the mappings

$$f(z) = R_{-1+2/k}(2/k, \ldots, 2/k; z - x_1, \ldots, z - x_{k-1}),$$

$$g(z) = R_{-1}(2/k, \ldots, 2/k; z - x_1, \ldots, z - x_k).$$

See Example 8.2-2 and Exercise 8.2-1 for the case $k = 4$, and Exercise 8.2-3 for $k = 3$.

8.2-3 Let $x_1, x_2 \in \mathbb{R}$ and assume $x_1 > x_2$. Denote by $z(u)$ the mapping inverse to

$$u(z) = R_{-1/3}(\tfrac{2}{3}, \tfrac{2}{3}; z - x_1, z - x_2).$$

Show that $z(u)$ is real on three families of parallel straight lines which divide the u plane into equilateral triangles. Show further that $z(u)$ is doubly periodic, that the smallest absolute value of any period is $M = \sqrt{3}u(x_1)$ [cf. Exercise 8.3-6], and that the periods of absolute value M form the vertices of a regular hexagon. [The function $z(u)$ is algebraically related to what is called the equianharmonic case of the Weierstrass function.]

8.3-1 Let $t \in \mathbb{C}$ and $\frac{1}{2} + \beta - t \in U$. Show that

$$R_{2t}\left(2\beta, \frac{1}{2} - \beta - t; \frac{1}{2}, 1\right) = \frac{\pi^{1/2}\Gamma(\frac{1}{2} + \beta - t)}{\Gamma(\frac{1}{2} + \beta)\Gamma(\frac{1}{2} - t)}.$$

8.3-2 Let $\alpha \in \mathbb{C}$ and $1 - \alpha + 2\beta \in U$. Show that

$$R_{-\alpha}(1 - \alpha, 2\beta; 1, 2) = \frac{2^{-2\beta}\pi^{1/2}\Gamma(1 - \alpha + 2\beta)}{\Gamma(\frac{1}{2} + \beta)\Gamma(1 - \alpha + \beta)}.$$

8.3-3 Let $a \in \mathbb{C}$ and $1 + a - b \in U$. Prove Kummer's theorem,

$${}_2F_1(a, b; 1 + a - b; -1) = \frac{2^{-a}\pi^{1/2}\Gamma(1 + a - b)}{\Gamma(1 - b + \frac{1}{2}a)\Gamma(\frac{1}{2} + \frac{1}{2}a)}.$$

This is the value of the ${}_2F_1$ function whether the ${}_2F_1$ series converges or not.

8.3-4 Let $a \in \mathbb{C}$ and $2c \in U$. Prove the formula (due to Kummer),

$${}_2F_1\left(2a, 1 - 2a; 2c; \frac{1}{2}\right) = \frac{\Gamma(c)\Gamma(c + \frac{1}{2})}{\Gamma(c + a)\Gamma(c - a + \frac{1}{2})}.$$

8.3-5 Let $x \in U$ and $y \in \mathbb{C}_>$. Prove the expansion (cf. Exercise 4.2-13),

$$\mathrm{B}(x, y) = \sum_{n=0}^{\infty} \frac{(1 - y, n)}{(x + n)n!}.$$

8.3-6 In Exercise 8.2-3 the least absolute value of a period was denoted by M. Show that

$$M = (2\pi)^{-1}[\Gamma(\tfrac{1}{3})]^3(x_1 - x_2)^{-1/3}.$$

8.3-7 Show that the arc length s, measured from the origin to a point with plane polar coordinates (r, θ) on the lemniscate $r^2 = \cos 2\theta$, is

$$s = \int_0^r (1 - t^4)^{-1/2}\, dt = R_F(r^{-2} + 1, r^{-2}, r^{-2} - 1),$$

the perimeter of the lemniscate being $P = 2\pi R_K(1, 2) = 5.24412\ldots$ (see Exercises 6.10-2 and 4.2-8). Show further that r^{-2} is the restriction to the real axis of a doubly periodic function of s with periods $P/2$ and $iP/2$. If r is considered to change sign whenever the point (r, θ) passes through the origin while describing the lemniscate, then the period of r is twice the period of r^{-2}. As a function of s, r is qualitatively similar to the sine function but has period $P = 2\pi R_K(1, 2)$ instead of 2π. It is called the lemniscatic sine function and was first studied by Gauss in 1797 [see Todd (1975)].

8.3-8 Let x, $t \in \mathbb{R}$ and c, $p \in \mathbb{R}_>$. Show that the solution of the differential equation $|dx/dt|^p + |x|^p = c^p$, with initial condition $dx/dt = c$ at $t = 0$, is

$$t = xR_{-1/p}(1/p, 1; c^p - |x|^p, c^p), \qquad |x| < c.$$

If $p > 1$, show that the solution is periodic with period $4(\pi/p)\csc(\pi/p)$.

8.3-9 Let $t \in \mathbb{C}$, $\beta + \frac{1}{2} \in U$, and x, $-x \in \mathbb{C}_0$ [whence $|\mathrm{ph}(-x^2)| = |\mathrm{ph}\, x + \mathrm{ph}(-x)| < \pi$]. Show that

$$R_{2t}(\beta, \beta; x, -x) = \frac{\pi^{1/2}\Gamma(\beta + \frac{1}{2})}{\Gamma(\frac{1}{2} - t)\Gamma(\beta + \frac{1}{2} + t)} (-x^2)^t.$$

(If $2t \in \mathbb{N}$, this reduces to Theorem 6.9-1.)

8.3-10 Let a, b, c, $x \in \mathbb{C}$ and d, $e \in U$, and define $s = d + e - a - b - c$. Assume $|x| \le 1$ and $s \in \mathbb{C}_>$. Show that the sum of the series ${}_3F_2(a, b, c; d, e; x)$ is continuous in x on the closed unit disk. Assume further that c, $e - c$, $d - a - b \in \mathbb{C}_>$ and show that

$${}_3F_2(a, b, c; d, e; x) = \int_0^1 {}_2F_1(a, b; d; ux)\, d\mu_{(c,\, e-c)}(u).$$

8.3-11 Let a, b, c, d, $e \in \mathbb{C}$ and define $s = d + e - a - b - c$. Assume $d \in U$ and c, $e - c$, $d - a - b \in \mathbb{C}_>$. Show that

$$B(c, e - c)\, {}_3F_2(a, b, c; d, e; 1) = B(c, s)\, {}_3F_2(d - a, d - b, c; d, s + c; 1).$$

8.3-12 Let a, b, c, d, $e \in \mathbb{C}$. Define $s = d + e - a - b - c$ and assume a, c, $d - a$, $e - c$, $d - a - b$, $e - a - c \in \mathbb{C}_>$. Show that

$$\frac{\Gamma(a)}{\Gamma(d)\Gamma(e)}\, {}_3F_2(a, b, c; d, e; 1) = \frac{\Gamma(s)}{\Gamma(s + b)\Gamma(s + c)}\, {}_3F_2(s, d - a, e - a; s + b, s + c; 1).$$

(It can be shown by analytic continuation that the conditions a, $s \in \mathbb{C}_>$ suffice.)

8.4-1 Show that the coefficient A_n in (8.4-2) is a multiple of $E_n(z)$ if $b = (\beta, \beta, \ldots, \beta)$.

8.4-2 Let $m \in \mathbb{N}^k$, $b - m \in U_k$, and $z \in \mathbb{C}_0{}^k$. Let $M = \sum_{i=1}^k m_i$ and a, $a' - M \in U$, and assume $a + a' = \sum_{i=1}^k b_i$. Prove the following generalization of (5.9-8):

$$\mathrm{B}(a, a' - M)R_{-a}(b - m, z) = \sum_{N=0}^{M} \binom{M}{N} R_N(-m, z)\mathrm{B}(a + N, a' - N)R_{-a-N}(b, z).$$

8.5-1 Let $x, y, z \in \mathbb{C}_0$ and $x^{1/2}, y^{1/2}, z^{1/2} \in \mathbb{C}_>$. Show that

$$R_{-1}\left(\frac{1}{2}, \frac{1}{2}, 1; x, y, z\right) = \frac{2}{(z - x)^{1/2}(z - y)^{1/2}} \log \frac{z^{1/2}[(z - x)^{1/2} + (z - y)^{1/2}]}{y^{1/2}(z - x)^{1/2} + x^{1/2}(z - y)^{1/2}}$$

$$= \frac{2}{(z - x)^{1/2}(z - y)^{1/2}} \log \frac{z + x^{1/2}y^{1/2} + (z - x)^{1/2}(z - y)^{1/2}}{z^{1/2}(x^{1/2} + y^{1/2})}.$$

This result will be used in Section 9.2. For another form of the right side see Exercise 9.8-5.

8.5-2 Let $w, x, y, z \in \mathbb{C}_0$ and assume that x, y, z are not in the closed convex hull of w and the negative real axis. Show that

$$R_{-1}(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; w, x, y, z) = 2[(x - w)(y - w)(z - w)]^{-1/2}$$

$$\cdot \left[R_F\left(\frac{1}{x - w}, \frac{1}{y - w}, \frac{1}{z - w}\right) - w^{1/2}R_F\left(\frac{x}{x - w}, \frac{y}{y - w}, \frac{z}{z - w}\right)\right].$$

For a simpler form of the right side see (9.8-4).